ZEPNEST

Tech-Enabled Home Services Platform

SOFTWARE REQUIREMENTS SPECIFICATION

Partner Application — Phase 1

Vendor & Technician Roles — Android & iOS

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification defines the complete functional and non-functional requirements for the Zepnest Partner Application — Phase 1. The Partner App serves two distinct user roles within a single application: Vendor and Technician. It is the operational backbone of the Zepnest platform — enabling job dispatch, execution, scope management, payment collection, and real-time status tracking.

1.2 Scope

The Zepnest Partner App (Phase 1) covers:

- Role-based login — Vendor and Technician roles within one app
- Vendor dashboard: technician list + live map tracking, job management, performance overview
- Technician job flows for all 3 job types: Rate Card Based, Service Based, High Ticket
- Scope change and revised pricing flow with customer approval via Customer App
- Rescheduling flow initiated by technician
- Payment collection via UPI QR and Cash confirmation
- Quote preparation for high-ticket inspection visits
- Earnings dashboard and performance metrics
- Notifications and support

1.3 Definitions & Abbreviations

Term	Definition
Vendor	A registered service partner business — owns and manages one or more technicians on the Zepnest platform
Technician	A field worker employed or contracted by the Vendor — executes jobs at customer premises
Job	A confirmed customer booking dispatched to a Vendor/Technician via the Zepnest platform
Ticket	Interchangeable with Job — the unit of work assigned and tracked in the system
Job Type 1	Rate Card Based — time-slot priced job (60/90/180 min) — Electrical, Plumbing
Job Type 2	Service Based — fixed price per service item — AC, Home Appliances
Job Type 3	High Ticket — inspection visit for Rental Turnaround, New Home Setup, Painting, Property Management
Scope Change	A situation where the job requires additional work or time beyond the original booking
Revised Quote	An updated price/scope sent to customer for approval when scope change occurs
OTP Confirmation	4-digit code from customer confirming job completion — required before job can be closed
UPI QR	A dynamic QR code generated on technician's app — customer scans with their UPI app to pay
COD	Cash on Delivery — cash given by customer to technician on completion
Vendor Dashboard	The Vendor role's home screen — shows all technicians, their status, and active jobs
Paired Job	AC Installation job requiring 2 technicians simultaneously

FR	Functional Requirement
NFR	Non-Functional Requirement

1.4 Role Overview

Role	Who	Primary Responsibility	Access Level
Vendor	Business owner / manager	Monitor technicians, manage jobs, view earnings, onboard team	Full dashboard + all technician views
Technician	Field worker	Execute assigned jobs, collect payment, update job status	Own jobs only — no access to other technicians' data

One app. Two roles. The login determines which dashboard the user sees. A vendor can also be a technician if they work in the field themselves.

2. OVERALL DESCRIPTION

2.1 System Architecture

The Partner App connects to the same Zepnest backend API as the Customer App and Admin Dashboard. Job dispatch, status updates, payment confirmation, and scope changes all flow through the central API — ensuring real-time synchronisation across all three platforms.

2.2 Operating Environment

Platform	Android primary (API 21+), iOS secondary — React Native or Flutter
Target Device	Low-to-mid range Android (₹8,000–₹15,000) — must perform flawlessly
Connectivity	Offline capability for core job functions — sync on reconnection
Location	GPS required — used for technician tracking and job navigation
Camera	Required — before/after photos, UPI QR display, document capture
Authentication	Mobile OTP — same mechanism as Customer App
Session	JWT — 30-day validity with silent refresh
Maps	Google Maps SDK — live map view for vendor + navigation for technician
Language	English only — Phase 1

2.3 Constraints & Assumptions

- Maximum 1 active job at a time per technician — next job visible but locked until current job is closed
- Technician cannot modify job type or service category — can only update scope, time, and price within allowed flow
- All price changes require customer approval via Customer App before work proceeds
- Vendor can only manage technicians who are formally linked to their Vendor account
- GPS tracking is active only during working hours (9am–7pm) and only when technician has an active/assigned job
- COD maximum ₹5,000 — enforced by system, not technician
- Technician app must function for core job actions (mark arrived, take photos, close job) with 2G connectivity

3. AUTHENTICATION & ROLE MANAGEMENT

3.1 Login Flow

Both Vendor and Technician use the same login screen. Role is determined by the mobile number registered in the Zepnest system — no role selection dropdown needed on login screen.

1	**Splash Screen** Zepnest Partner App logo. Check existing JWT session → if valid, navigate to role-appropriate dashboard.
2	**Login Screen** 'Enter your mobile number' field + 'Send OTP' button. Clean, minimal screen. No username/password.
3	**OTP Verification** 6-digit OTP via SMS. 60-second countdown. Resend option. Auto-read via SMS Retriever API on Android.
4	**Role Detection** System detects role from mobile number. Vendor → Vendor Dashboard. Technician → Technician Job Dashboard. If number not registered → 'Contact Zepnest to register as a partner.'
5	**First Login** If first login for technician: show induction checklist status, pending training modules, zone assignment.

3.2 Dual Role — Vendor Who Is Also a Technician

If a vendor also works as a field technician (common in small operations), their account has both roles. On login — app detects dual role and presents a role selection screen:

- 'I'm managing today' → Vendor Dashboard
- 'I'm working in the field today' → Technician Job Dashboard
- Role can be switched at any time from the profile menu

3.3 Authentication Requirements

FR-AUTH-001	System shall determine user role from registered mobile number. No manual role selection on login. Priority: P1
FR-AUTH-002	If mobile number has both Vendor and Technician roles, system shall present a role selection screen after OTP verification. Priority: P1
FR-AUTH-003	Unregistered mobile number login attempt shall display: 'This number is not registered as a Zepnest Partner. Contact Zepnest to get started.' Priority: P1
FR-AUTH-004	JWT session valid for 30 days. App shall silently refresh token without requiring re-login. Priority: P1
FR-AUTH-005	Logout shall clear local session, cached job data, and navigate to login screen. Priority: P1

4. VENDOR DASHBOARD

The Vendor Dashboard is the command centre. The vendor sees everything — where every technician is, what every job status is, and how the business is performing.

4.1 Vendor Home Screen Layout

The Vendor home screen has three main components accessible via tabs at the top:

- Tab 1 — My Team (default): Technician list + live status
- Tab 2 — Map View: Live map showing all technicians' locations
- Tab 3 — Jobs: All active and scheduled jobs across all technicians

4.2 Tab 1 — My Team (Technician List View)

4.2.1 Technician Status Cards

Each technician shown as a card with:

- Technician photo (or initials avatar if no photo)
- Technician name and primary skill category
- Status badge — colour coded:
  - 🟢 Available — no active job, online
  - 🟡 En Route — travelling to job
  - 🔵 On Job — currently working at customer premises
  - 🔴 Offline — not available / app inactive
  - ⚪ Off Duty — outside working hours
- Current job name and address (if on job or en route)
- Next job scheduled (if applicable)
- Today's completed job count
- Today's earnings (visible to Vendor only — not shown to technician in Vendor's view publicly)
- 'View Details' button → opens Technician Detail screen

4.2.2 Technician Detail Screen (Vendor View)

- Full profile: name, mobile, skill categories, zone, tier status, BGV status
- Today's job schedule: all assigned jobs in sequence with time slots and status
- Live location on mini-map (active during working hours when job is assigned)
- 30-day rating and recent customer reviews
- Earnings this week and this month
- Performance metrics: acceptance rate, on-time arrival %, average rating
- 'Reassign Job' option — vendor can reassign any pending/assigned job to another technician
- 'Contact Technician' — masked phone call

4.3 Tab 2 — Map View

4.3.1 Live Map Features

- Google Maps base layer showing Zepnest's serviceable zone (600048 highlighted)
- Each technician shown as a pin on the map with their initials
- Pin colour matches status: green (available), yellow (en route), blue (on job), red (offline)
- Tapping a technician pin opens a bottom sheet with: name, current status, current job address, next job, quick contact button
- Customer job locations shown as home icons — tapping shows job details
- Map auto-refreshes technician positions every 30 seconds
- 'Fit All' button — zooms map to show all technicians simultaneously

- Filter by status: show only Available / On Job / En Route technicians

4.3.2 Map — GPS Accuracy Note

GPS tracking is active only for technicians with an assigned job during working hours (9am–7pm). Offline or off-duty technicians do not appear on map.

4.4 Tab 3 — Jobs (All Jobs View)

4.4.1 Jobs List

- Filter tabs: Today / Tomorrow / This Week
- Each job card shows: job ID, service type (Type 1/2/3), customer address, assigned technician, time slot, status badge, amount
- Sort by: Time (default), Status, Technician, Amount
- Search by: Job ID, customer address, technician name
- Tap any job → Job Detail screen

4.4.2 Job Detail Screen (Vendor View)

- Complete job information: type, services, customer first name, address, slot, assigned technician
- Real-time job status with timestamp log: Assigned → En Route → Arrived → In Progress → Completed
- Scope change history: if technician raised scope change, vendor can see original scope, revised scope, customer approval status
- Payment status: UPI paid / COD pending / COD collected
- Before and after photos uploaded by technician (viewable by vendor)
- Customer rating (visible after submission)
- 'Reassign Technician' button (only visible if job status is Assigned — before technician is En Route)
- 'Escalate to Zepnest Ops' button — for any job issues requiring admin intervention

4.5 Vendor — Additional Screens

4.5.1 Earnings Screen

- This Week / This Month / Custom Date Range toggle
- Total revenue (gross customer payments)
- Platform fee deducted (Zepnest's share)
- MRG top-up received (if applicable — during MRG period)
- MRG recovery deducted (if applicable)
- Net payout to vendor
- Per-technician earnings breakdown
- Per-job earnings list with drill-down
- Download weekly settlement statement (PDF)

4.5.2 Performance Screen

- Team average rating (30-day rolling)
- Total jobs completed this month
- On-time arrival rate (%)
- Acceptance rate (%)
- Scope change rate (%) — how often jobs go beyond original scope
- Top performing technician of the month
- Complaints received and resolution status

4.6 Vendor Functional Requirements

FR-VEND-001	Vendor dashboard shall display real-time status of all linked technicians — list view and map view. Status refresh maximum 30-second interval. Priority: P1
FR-VEND-002	Vendor shall be able to view complete job details for all jobs assigned to their technicians — including scope changes and payment status. Priority: P1
FR-VEND-003	Vendor shall be able to reassign a job to another technician only when job status is 'Assigned' (before technician has marked 'En Route'). Priority: P1
FR-VEND-004	Vendor shall see technician GPS location on map only during working hours and only when technician has an active or assigned job. Priority: P1
FR-VEND-005	Vendor shall receive push notification for: new job assigned to their team, scope change raised by technician, job completed, customer complaint raised. Priority: P1
FR-VEND-006	Vendor earnings screen shall match Zepnest weekly settlement statement exactly — same figures, no discrepancies. Priority: P1

5. TECHNICIAN — HOME SCREEN & JOB DASHBOARD

5.1 Technician Home Screen Layout

The Technician Home Screen is designed for one-handed use in the field. All critical actions are within thumb reach. Screen sections:

- Header: Zepnest Partner logo | Technician name | Online/Offline toggle | Notification bell
- Status Bar: Current status badge + active job name (if any)
- Today's Earnings: Large number display — updates in real time after each job
- Today's Jobs: Vertical list of all jobs assigned for today in time order
- Quick Actions: 'Navigate to Next Job' button (active when job is Assigned)

5.2 Online / Offline Toggle

- When Online: technician is visible to Vendor on map, eligible to receive new job dispatches
- When Offline: technician disappears from Vendor map, no new jobs dispatched
- Toggle requires confirmation: 'Going offline will pause new job assignments. Confirm?'
- System auto-sets Offline after working hours (7pm) and auto-Online at start of working hours (9am) — can be overridden

5.3 Today's Jobs List

All jobs assigned for the day shown in chronological order by slot time:

- Job card shows: job type badge (Type 1/2/3), time slot, customer address (locality only — no flat number until En Route), service name, amount
- Active job highlighted in blue border
- Upcoming jobs shown with countdown: 'Next job in 45 min'
- Completed jobs shown with green tick
- Each job card tappable → Job Detail screen

5.4 Technician Status Flow — Across All Job Types

Status	Triggered By	Customer App Update	Vendor Dashboard Update
Assigned	Zepnest Admin / system dispatch	'Technician assigned'	Technician shows as 'Assigned'
Accepted	Technician taps 'Accept Job'	No change	Vendor notified job accepted
En Route	Technician taps 'On My Way'	'Technician is on the way'	Map shows technician moving
Arrived	Technician taps 'I've Arrived'	'Technician has arrived'	Status updated on dashboard
In Progress	Technician taps 'Start Work'	'Work in progress'	Timer visible on vendor view
Scope Change	Technician raises scope change	Customer gets approval notification	Vendor notified of scope change
Completed	Customer OTP entered + job closed	'Job completed' + rating prompt	Job marked complete + earnings updated

Rescheduled	Technician/Vendor initiates reschedule	Customer notified of new slot	New slot visible on dashboard

6. JOB TYPE 1 — RATE CARD BASED JOBS (ELECTRICAL / PLUMBING)

Time-based pricing. Customer booked a 60, 90, or 180-minute slot. Technician executes within that time and documents all work done.

6.1 Job Type 1 — Technician Flow

1	**Job Notification & Acceptance** New job push notification received. Job card shows: service (Electrical/Plumbing/Both), time slot, customer locality, package booked (60/90/180 min), amount. 'Accept' or 'Decline' buttons. 15-minute acceptance window.
2	**Pre-Job Checklist** 30 minutes before slot: checklist notification. App shows tools required for job type. Technician confirms readiness. Any tool gap flagged to Vendor.
3	**Navigate to Customer** Tap 'On My Way' — triggers GPS tracking + customer notification. 'Open in Google Maps' button shows full address. Maps app opens with navigation.
4	**Arrival** On reaching customer premises: tap 'I've Arrived'. Full customer address now revealed. Customer notified. Timer countdown visible showing booked slot duration.
5	**Before Photo** Mandatory: tap 'Take Before Photo'. In-app camera opens. Minimum 2 photos required (affected areas). GPS-tagged + timestamped. Cannot proceed to 'Start Work' without photos.
6	**Start Work** Tap 'Start Work' — job timer begins counting down from booked duration (60/90/180 min). Customer sees 'Work in Progress' in their app. Timer visible to technician throughout.
7	**Work Summary Entry** During or after work: technician fills Work Summary. Two components: (A) Task Checklist — select all tasks performed from predefined list; (B) Free Text Notes — additional notes, materials used, observations.
8	**After Photo** Mandatory: tap 'Take After Photo'. Minimum 2 photos. GPS-tagged + timestamped. Cannot proceed to job closure without.
9	**Customer OTP** Tap 'Request Customer OTP'. Customer receives 4-digit OTP in their app + SMS. Technician enters OTP in Partner App. OTP valid for 10 minutes.
10	**Payment Collection** After OTP confirmed: payment screen. Show UPI QR (dynamic — generated for this job amount) + 'Cash Collected' option. Customer scans QR with their UPI app and pays. Or technician marks 'Cash Collected'.
11	**Job Closed**

	After payment confirmed: job status = Completed. Earnings updated on dashboard. Rating prompt triggered on Customer App. Next job unlocked.

6.2 Work Summary — Predefined Task List (Type 1)

Electrical Tasks:

- Fan installation / replacement
- Light fitting / bulb replacement
- Switch / socket replacement
- MCB / circuit breaker replacement
- Fan regulator fitting
- Doorbell installation
- Extension board fitting
- Earthing check and repair
- Wiring inspection and repair
- DB box work
- Other (free text)

Plumbing Tasks:

- Tap / faucet repair or replacement
- Pipe leak repair
- Drain / sink unclog
- Toilet flush repair
- Shower / mixer fitting
- Water tank valve repair
- Geyser connection check
- Water pressure issue fix
- Concealed pipe inspection
- Other (free text)

6.3 Timer Management

- Technician sees live countdown timer throughout the job
- At 10 minutes remaining: yellow warning — 'Your 60-min slot ends in 10 minutes'
- At 5 minutes remaining: red warning — 'Slot ending soon'
- Timer expiry triggers Scope Change flow automatically if job not closed (see Section 8)

6.4 Job Type 1 Functional Requirements

FR-T1-001	Before photos (minimum 2) are mandatory before 'Start Work' can be tapped. System shall block progression without photos. Priority: P1
FR-T1-002	Job timer shall begin exactly when technician taps 'Start Work' and be visible throughout the job in hours:minutes:seconds format. Priority: P1
FR-T1-003	After photos (minimum 2) are mandatory before customer OTP can be requested. System shall block OTP request without after photos. Priority: P1

FR-T1-004	Customer OTP shall be 4 digits, valid for 10 minutes. Maximum 3 incorrect attempts before OTP regeneration required. Priority: P1
FR-T1-005	Work summary (task checklist selection) is mandatory before job can be closed. Free text notes are optional. Priority: P1
FR-T1-006	Timer alert shall display at 10 minutes and 5 minutes remaining. At timer expiry with job still open, system shall automatically prompt scope change flow. Priority: P1

7. JOB TYPE 2 — SERVICE BASED JOBS (AC / HOME APPLIANCES)

Fixed price per service item. Customer booked specific services from the catalogue. Technician completes those exact services and documents with photos.

7.1 Job Type 2 — Technician Flow

The Job Type 2 flow is similar to Type 1 with one key difference: instead of a timer, the technician works from a fixed service checklist based on what the customer booked.

1 **Job Notification & Acceptance**
Job card shows: services booked (e.g., 'AC Deep Clean + AC Gas Refill'), customer locality, time slot, total amount. Accept within 15 minutes.

2 **Navigate to Customer**
'On My Way' tap → GPS tracking + customer notification + Google Maps navigation.

3 **Arrival & Before Photo**
'I've Arrived' → full address revealed. Mandatory: minimum 2 before photos of relevant appliance/area. Job-specific: AC photos, appliance condition photos.

4 **Start Work**
Tap 'Start Work'. Service checklist appears — shows all booked services as checkboxes. Technician works through each item.

5 **Service Checklist Completion**
As each service is completed, technician checks it off. Checked items are locked — cannot uncheck. System tracks completion percentage.

6 **Service-Specific Data Entry**
Per service type, additional data fields appear:

Service	Required Data Entry
AC Service	Pre and post cleaning photos, filter condition (Good/Replace), coil condition, any leaks found (Y/N)
AC Gas Refill	Gas type (R22/R32/R410A), kg added, pre and post pressure readings
AC Installation	Indoor and outdoor unit photos, pipe length used, electrical connection type
Appliance Installation	Installation photos, test run result (Pass/Fail), serial number captured
Appliance Repair	Fault description, parts replaced (from parts list), repair photos
Inverter Battery Water	Before/after cell photos, water level before and after, distilled water litres added
Water Purifier Install	TDS reading before and after, filter type installed, connection photos

7 **Parts Used Log**
If any parts were used during the job, technician logs: part name, quantity, unit cost. These are added to the customer invoice as actual-cost line items.

8 **After Photos**

	Minimum 2 after photos. Appliance-specific: installed unit, work area, test results.
9	**Customer OTP + Payment** Same as Type 1 — OTP confirmation → UPI QR or Cash collection → Job closed.

7.2 Job Type 2 Functional Requirements

FR-T2-001	Service checklist shall be pre-populated based on services booked by customer — technician cannot add or remove checklist items. Priority: P1
FR-T2-002	All items on the service checklist must be checked before after photos can be taken. Uncompleted items block progression. Priority: P1
FR-T2-003	Service-specific data fields (gas pressure readings, TDS readings, serial numbers) shall be mandatory for relevant service types. Priority: P1
FR-T2-004	Parts used log shall generate a separate line item on the customer invoice at actual cost — no platform fee applied on parts. Priority: P1
FR-T2-005	For AC gas refill jobs, gas type and quantity fields are mandatory — cannot close job without this data. Priority: P1

8. JOB TYPE 3 — HIGH TICKET SERVICES (INSPECTION VISIT)

This is an inspection and quote visit — not execution. Technician visits, assesses scope, prepares and presents a detailed quote to the customer. If approved, a new execution job is created.

8.1 High Ticket Services in Scope

- Rental Turnaround — full unit refurbishment assessment
- New Home Setup — post-builder handover setup assessment
- Painting — interior/exterior scope and surface assessment
- Property Management — free or paid plan onboarding visit
- Any job estimated above ₹2,000 or requiring more than 2 hours

8.2 Job Type 3 — Technician Flow

1	**Job Notification & Acceptance** Job card shows: 'INSPECTION VISIT', service type (Rental Turnaround / New Home Setup / Painting etc.), customer address, time slot, ₹149 visit fee. Accept within 15 minutes.
2	**Navigate & Arrive** 'On My Way' → navigate → 'I've Arrived' → full address revealed.
3	**Start Visit** Tap 'Start Inspection Visit'. Visit timer begins — for internal tracking only (not shown to customer). Checklist of inspection areas appears based on service type.
4	**Scope Gathering — Photos** Technician photographs all relevant areas per inspection checklist. Photos are tagged by area (Living Room, Kitchen, Bathroom, Electrical panel, etc.).
5	**Scope Gathering — Observations** Per area: free text observations. 'What needs to be done?' — technician describes scope in simple language.
6	**Quote Preparation** Technician opens Quote Builder:

Quote Builder has two modes:

- Mode A — Structured: Select service items from predefined list, enter quantity, system calculates subtotal per item and grand total
- Mode B — Free Text + Total: Type custom description, enter total amount manually
- Both modes can be combined in one quote — structured items + additional free text items

Quote Builder — Predefined Item Library (Sample):

Category	Items Available
Painting — Labour	Per sq ft rate (basic/texture/waterproof) — quantity = sq ft
Electrical	Fan installation, light fitting, switch replacement, wiring — per unit
Plumbing	Tap replacement, pipe repair, drain unclog — per unit

Deep Cleaning	Per BHK size (1BHK/2BHK/3BHK)
AC Service	Basic clean, deep clean — per unit
Carpentry	Door repair, furniture assembly — per item
Miscellaneous	Free text item + manual price entry

7	**Quote Review** Technician reviews complete quote: itemised list, subtotal, Zepnest platform fee (pre-calculated, not shown to customer), total payable by customer. Edit option available before presenting.

8	**Present Quote to Customer** Technician taps 'Present Quote to Customer'. Two options:

- Option A — Show on Screen: Quote displayed on technician's phone — customer reads and reviews in person
- Option B — Send to Customer App: Quote pushed to customer's Zepnest app — customer receives push notification 'Your quote is ready. View and approve.'

For in-person presentation (Option A), technician walks customer through each line item verbally. Customer can ask questions. After discussion, technician sends quote to Customer App for formal digital approval.

9	**Customer Approval — Via Customer App** Customer opens their Zepnest app. Views itemised quote with total. 'Approve & Book' or 'Decline' buttons. If approved → new execution job automatically created and scheduled. ₹149 visit fee credited to new job total.

10	**Post-Approval — New Job Created** Technician receives notification: 'Customer approved your quote! Execution job [ID] has been created.' New job appears in technician's upcoming jobs. Scheduling handled by Zepnest ops.

11	**Quote Declined** If customer declines: technician notified. Visit job closed. ₹149 collected (not refundable on decline). Technician can note decline reason for ops review.

8.3 Inspection Checklist — By Service Type

Service Type	Inspection Areas	Key Observations
Rental Turnaround	All rooms, kitchen, bathrooms, balcony, electrical panel, plumbing points	Paint condition, damage, electrical faults, plumbing issues, cleaning scope, fixture repairs needed
New Home Setup	All rooms, electrical points, plumbing connections, AC positions	Fan/light points available, AC positions, appliances to install, wiring completeness
Painting	All interior/exterior walls, ceiling, doors, windows, special surfaces	Surface condition, sq ft measurement, texture requirements, waterproofing areas
Property Management	Full home walkthrough, all assets, garden/trees, inverter	Asset condition, warranty check, maintenance backlog, tree condition

8.4 Job Type 3 Functional Requirements

FR-T3-001	Quote Builder shall support both structured item selection and free text entry simultaneously in a single quote. Priority: P1
FR-T3-002	Predefined item library in Quote Builder shall be maintained by Zepnest Admin — updatable without app release. Priority: P2
FR-T3-003	Quote sent to Customer App shall be identical to what was prepared in Partner App — no data loss or formatting change. Priority: P1
FR-T3-004	On customer quote approval in Customer App, execution job shall be automatically created in the Zepnest system with all quote line items as the job scope. Priority: P1
FR-T3-005	₹149 inspection visit fee shall be automatically deducted from the total of the approved execution job. Priority: P1
FR-T3-006	Inspection photos tagged by area shall be stored and accessible in the Admin Dashboard and Vendor Dashboard — not just on the technician's device. Priority: P1

9. SCOPE CHANGE & REVISED PRICING FLOW

Scope change can be triggered in Job Type 1 (timer expiry or discovery) and Job Type 2 (additional items found). Customer must approve all price changes before work proceeds.

9.1 Scope Change Triggers

- Job Type 1: Timer expires before work is complete
- Job Type 1: Technician discovers additional issues during work requiring extra time or materials
- Job Type 2: Additional items found during service that are not in original booking
- Any job type: Material cost exceeds standard expectation and customer approval is needed

9.2 Scope Change Flow

1	**Technician Initiates** Technician taps 'Report Scope Change' button (visible on active job screen at all times). System pauses the job timer (Type 1) while scope change is being processed.
2	**Scope Change Form** Technician fills:

- Reason for scope change (dropdown): Additional work found / Job requires more time / Materials needed / Out of scope discovery / Other
- Description (free text): What was found, what additional work is needed — plain language
- Additional time needed (Type 1 only): 30 min / 60 min / 90 min / Custom
- Revised price: New total (for time extension: additional slot cost shown automatically)
- Photo evidence (optional but recommended): Photo of the issue requiring scope change

3	**Scope Change Preview** Technician reviews: original booking summary, scope change reason, revised scope, original price, additional amount, new total. 'Send to Customer' button.
4	**Sent to Customer** Customer receives push notification: 'Action Required: Your technician has flagged a scope change. Review and approve.' Customer App shows: original booking, what changed, original price, additional amount, new total. 'Approve' and 'Decline' buttons. 10-minute approval window.
5A	**Customer Approves** Technician notified: 'Customer approved. Resume work.' Timer resumes (Type 1) with additional time added. Work continues. Job closes at new total amount.
5B	**Customer Declines** Technician notified: 'Customer declined scope change.' Technician must complete original scope only — within original budget and time. Close job normally at original price.30
5C	**Approval Window Expires** If customer doesn't respond in 10 minutes: technician receives 'No response from customer' alert. Options: Try calling customer (masked call from app) / Escalate to Zepnest ops / Close job at original scope.

9.3 Scope Change Functional Requirements

FR-SC-001	Scope change form shall be accessible at any point during an active job — visible as a persistent button on the job screen. Priority: P1
FR-SC-002	Job timer (Type 1) shall automatically pause from the moment technician submits a scope change until customer approves or declines. Priority: P1
FR-SC-003	Customer approval window for scope change is 10 minutes. System shall alert technician at 2 minutes remaining. On expiry, technician receives options to call, escalate, or close at original scope. Priority: P1
FR-SC-004	Scope change history — all scope changes raised, customer response, and final outcome — shall be logged against the job and visible to Vendor and Zepnest Admin. Priority: P1
FR-SC-005	If customer approves scope change, the job's total amount in the payment system shall automatically update to the approved revised amount before payment collection. Priority: P1

10. RESCHEDULING FLOW

Technician can initiate a reschedule when a job cannot be completed in the current slot due to unforeseen circumstances. All reschedules require Zepnest ops approval.

10.1 Reschedule Trigger Conditions

- Job requires significantly more time than any available extension (multi-day work discovered)
- Customer not available / not at premises after arrival
- Critical material not available — must be sourced before work can proceed
- Emergency on-site safety issue preventing work
- Customer requests reschedule after technician has arrived

10.2 Reschedule Flow

1	**Technician Initiates** Tap 'Request Reschedule' on active job screen. Available only after status = Arrived (not before technician reaches customer).
2	**Reschedule Form** Technician fills: Reason for reschedule (dropdown + free text), Suggested new date (next 3 days), Suggested time slot, Photo evidence if applicable (e.g., photo of missing material, safety issue).
3	**Sent to Zepnest Ops** Reschedule request goes to Admin Dashboard for review — not directly to customer. Ops team reviews within 30 minutes during working hours.
4	**Ops Reviews & Approves** Zepnest ops confirms new slot availability, approves reschedule, and updates the booking system. Customer notified via push notification and SMS: 'Your [service] has been rescheduled to [new date] at [new time]. We apologise for the inconvenience.'
5	**Technician Notified** Technician receives: 'Reschedule approved. New slot: [date] [time].' Current job status changes to 'Rescheduled'. ₹149 visit fee for high-ticket jobs is not charged for rescheduled visit (Zepnest ops discretion).
6	**Ops Declines Reschedule** If ops declines (e.g., reason not valid): technician notified. Must either complete the job or escalate to ops for further resolution.

10.3 Reschedule Functional Requirements

FR-RESCH-001	Reschedule option shall only be available after technician has marked 'Arrived'. Not available before arrival. Priority: P1
FR-RESCH-002	All reschedule requests shall route to Zepnest Admin Dashboard — not directly to customer. Ops team is the approver. Priority: P1

FR-RESCH-003	Customer shall be notified of reschedule only after Zepnest ops has approved it — not when technician raises the request. Priority: P1

FR-RESCH-004	Reschedule reason, photo evidence, and ops decision shall be logged against the job permanently for quality review. Priority: P1

11. PAYMENT COLLECTION

11.1 Payment Methods — Technician Side

Method	Mechanism	When Available	Technician Action
UPI QR	Dynamic QR generated per job amount. Customer scans with their UPI app.	All jobs — primary method	Display QR on screen. Wait for payment confirmation from backend.
Cash	Customer pays cash to technician.	Jobs ≤ ₹5,000. COD selected by customer at booking.	Tap 'Mark Cash Collected'. Enter amount received. System confirms.

11.2 UPI QR Payment Flow

1 **Payment Screen Opens**
After customer OTP confirmed, payment screen automatically opens showing job total (including any approved scope change additions).

2 **Dynamic QR Generated**
App calls Razorpay API to generate a dynamic UPI QR code for the exact job amount. QR displayed full-screen on technician's phone. Amount shown prominently below QR.

3 **Customer Scans**
Technician holds phone for customer to scan QR. Customer scans with any UPI app (GPay, PhonePe, Paytm, etc.). Customer sees amount and 'Zepnest' as payee. Customer approves payment in their UPI app.

4 **Payment Confirmation**
Razorpay webhook confirms payment success to Zepnest backend. Partner App receives real-time confirmation: green tick + 'Payment Received ₹[amount]'. QR screen replaced with success screen.

5 **Payment Failure**
If customer's UPI payment fails: 'Payment failed. Please retry or pay cash.' Technician can regenerate QR or switch to cash collection. Maximum 3 QR payment attempts.

6 **Job Closed**
Payment confirmed → job automatically closes → earnings updated → rating prompt on Customer App.

11.3 Cash Collection Flow

1 **Cash Screen**
After OTP: 'Cash on Delivery' screen shows amount to collect. 'Mark Cash Collected' button.

2 **Amount Confirmation**
Technician taps 'Mark Cash Collected'. Confirmation dialog: 'Confirm ₹[amount] received in cash from customer?' Technician confirms.

3	**Job Closed** Cash marked → job closed → customer receives digital receipt via SMS → earnings logged.

11.4 Payment Functional Requirements

FR-PAY-001	Dynamic UPI QR shall be generated via Razorpay API for the exact job amount — including any scope change additions. Static/pre-generated QR codes are not permitted. Priority: P1
FR-PAY-002	QR payment confirmation shall arrive via Razorpay webhook in maximum 30 seconds. If no confirmation in 60 seconds, system shall prompt technician to ask customer to check their UPI app. Priority: P1
FR-PAY-003	Cash collection marking is permanent — cannot be undone by technician after confirmation. Disputes must go through Zepnest ops. Priority: P1
FR-PAY-004	Customer digital receipt shall be sent via SMS within 60 seconds of payment confirmation (UPI or Cash). Priority: P1
FR-PAY-005	Payment screen shall not allow job closure without either UPI payment confirmation or Cash collection confirmation. Priority: P1

12. TECHNICIAN — ADDITIONAL SCREENS

12.1 Earnings Screen

- Today's earnings — large display, real-time update after each job close
- This week / This month toggle
- Per-job earnings breakdown: job ID, service, amount, platform fee, payout, date
- Bonus earnings: performance bonuses, AMC conversion bonuses, referral bonuses
- Penalty deductions: SLA breaches, missing photos, no-show fees — itemised
- Net payout: what will be settled on Friday
- Settlement countdown: 'Next payment in X days — estimated ₹Y'
- Cumulative total since joining Zepnest

12.2 My Performance Screen

- Current 30-day rolling rating — star display + numerical
- Rating trend graph — last 90 days
- Individual job ratings — customer feedback per job (customer name masked)
- SLA metrics: acceptance rate (%), on-time arrival (%), jobs completed vs assigned
- Scope change rate — how often jobs require scope change
- Current tier: Associate / Certified Partner / Senior Partner / Team Lead
- Progress to next tier: conditions met and remaining
- Mystery audit results (if any)

12.3 Training Screen

- Mandatory training modules — must complete before first job is unlocked
- Module list: The Zepnest Way, App Usage, Customer Communication, Safety, Payment Collection
- Each module: short video (YouTube link) + 5-question quiz
- Completion status per module: Not Started / In Progress / Completed
- Overall induction progress percentage
- New modules added by Zepnest Admin appear here — push notification on new module
- Certificate of completion downloadable as PDF after all mandatory modules done

12.4 Profile Screen

- Technician photo (editable)
- Name, mobile, skill categories, assigned zone
- BGV status: Verified / Pending / Expired
- OEM certifications held (Daikin, Voltas, Kent, etc.)
- Joining date and Zepnest Partner ID
- Documents uploaded: Aadhar (status), address proof, skill certificates
- Vendor name (which vendor the technician is linked to)
- 'Report an Issue' — quick support access
- Logout

12.5 Notifications Screen

- In-app notification inbox — all notifications in reverse chronological order
- Filter tabs: All / Jobs / Payments / Performance
- Unread count badge on notification bell in header

- Notification types:
  - New job dispatched — with accept/decline quick action
  - Job reminder (30 min before slot)
  - Scope change customer response (approved/declined)
  - Payment confirmed
  - Customer rating received
  - Reschedule approved/declined by ops
  - Bonus earned
  - Penalty applied
  - Settlement processed
  - New training module available
  - Tier upgrade achieved

12.6 Help & Support Screen

- FAQ: top 15 questions for technicians — collapsible accordion
- 'Call Zepnest Ops' — direct line to operations team (business hours)
- 'WhatsApp Zepnest' — deep-link to support WhatsApp
- 'Report Job Issue' — raise issue against a specific job with description and photos
- 'Report Customer Issue' — flag customer misconduct or unsafe situation
- Emergency SOS — visible on active job screen — connects to ops immediately

13. NON-FUNCTIONAL REQUIREMENTS

13.1 Performance

NFR-PERF-001	App launch to job dashboard: maximum 3 seconds on 4G. Maximum 7 seconds on 2G. Priority: P1
NFR-PERF-002	UPI QR generation (Razorpay API call): maximum 3 seconds. If timeout, show retry option. Priority: P1
NFR-PERF-003	Photo capture and local storage: instantaneous. Photo upload to server: in background — job can proceed without waiting for upload. Sync on connectivity. Priority: P1
NFR-PERF-004	Map view in Vendor Dashboard: technician pins shall update position within 30 seconds of actual movement. Priority: P2
NFR-PERF-005	Scope change customer approval notification shall reach customer app within 10 seconds of technician submission. Priority: P1

13.2 Offline Capability

Function	Works Offline?	Sync Behaviour
View today's accepted jobs	YES	Pre-cached on job acceptance
Customer address (after En Route)	YES	Pre-cached on job acceptance
Take and store job photos	YES	Uploads when connectivity restored
Mark Arrived / Start Work / In Progress	YES	Syncs to server within 60 sec of connectivity
Work summary / task checklist entry	YES	Syncs on connectivity
Accept new job alerts	NO	Requires push notification connectivity
Scope change submission	NO	Requires connectivity — customer approval needed
UPI QR generation	NO	Requires Razorpay API call
Customer OTP verification	NO	Requires server-side OTP validation
Vendor map view	NO	Requires real-time GPS data

13.3 Security

NFR-SEC-001	Customer full address (flat number) shall only be revealed to technician after 'En Route' is marked — not at job acceptance. Priority: P1
NFR-SEC-002	Customer mobile number shall never be displayed to technician. All calls via masked proxy number. Priority: P1

NFR-SEC-003	All API communication via HTTPS TLS 1.2+. JWT stored in Android Keystore / iOS Keychain. Priority: P1
NFR-SEC-004	Technician GPS data retained for maximum 30 days — used only for Vendor dashboard and quality audits. Priority: P1
NFR-SEC-005	UPI QR codes are dynamic and single-use — generated per job and expire after payment or 30 minutes. Priority: P1

13.4 Device & Usability

NFR-USE-001	App must function on ₹8,000 Android devices in direct sunlight with 2G connectivity for all critical job actions. Priority: P1
NFR-USE-002	Minimum touch target size: 48x48px. Large action buttons (Start Work, Mark Arrived, Take Photo) shall be full-width — minimum 56px height. Priority: P1
NFR-USE-003	Camera interface for photo capture must open within 2 seconds. Screen shall not lock during active job. Priority: P1
NFR-USE-004	App installed size: maximum 50MB. No background data usage beyond GPS tracking and push notifications during non-active periods. Priority: P2

14. OPEN POINTS & PHASE 2 SCOPE

14.1 Open Points — Decisions Required Before Development

#	Open Point	Decision Needed From	Impact
1	Razorpay merchant account — dynamic QR API access and webhook configuration	Founder	UPI payment collection blocked without this
2	GPS tracking — confirm if Google Maps Platform pricing is acceptable for Vendor map view refresh frequency	Dev Team	Map view cost planning
3	Predefined quote item library for Type 3 jobs — who maintains it and initial item list	Founder + Ops	Quote Builder cannot be built without initial library
4	Masked phone call proxy service — Exotel or similar — account setup	Founder	Technician ↔ customer calling blocked
5	Admin Dashboard spec — how ops approves reschedules, assigns jobs, manages scope changes	Dev Team	All ops-dependent flows blocked
6	Vendor onboarding flow — how vendors are linked to technicians in the backend	Dev Team	Multi-technician vendor model blocked
7	Photo storage — S3 or equivalent — retention policy and storage cost approval	Dev Team / Founder	Photo upload and report generation blocked

14.2 Explicitly Out of Scope — Phase 1

- In-app chat between technician and customer — Phase 2
- Technician to technician job transfer (peer transfer) — Phase 2
- Technician-initiated new job creation (walk-in customers) — Phase 2
- Inventory management (parts and materials tracking) — Phase 2
- Tamil language support — Phase 2
- Vendor-to-vendor job referral — Phase 2
- Automated job dispatch (AI-based) — Phase 3

14.3 Phase 2 — Planned Features

- In-app chat (technician ↔ customer, technician ↔ vendor, technician ↔ ops)
- Tamil language toggle
- Inventory and parts management module for vendors
- Automated smart dispatch — system assigns jobs based on proximity, skill, and rating
- Technician earnings advance request (salary-on-demand)
- Multi-zone vendor support — vendor managing technicians across multiple pincodes
- Technician peer review — team leads rate junior technicians
- Integration with OEM service management systems (Daikin, Voltas)